

Mode Coverage in Boltzmann Generators via Log-Ratio Variation

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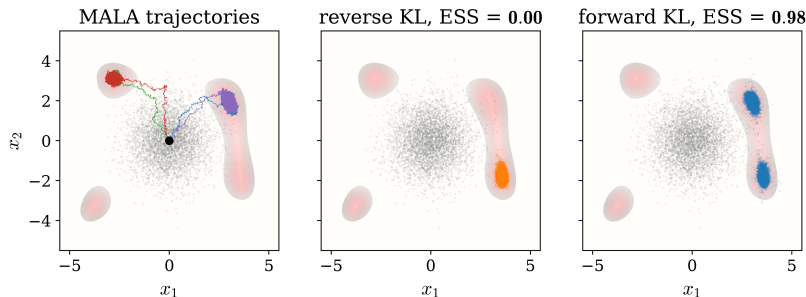
The Sampling Problem, and Where It Fails

- Given a potential U on $\mathbb{R}^d$, draw samples from

$$\pi(x) \propto e^{-U(x)},$$

with access to U and ∇U alone.

- U is multimodal: the mass splits among basins separated by barriers, and the quantity wanted is a ratio of masses between basins.
- Mode collapse**: the sampler occupies some basins and never reaches the rest. Nothing it reports shows the absence.



Boltzmann Generators: a Global Transport

One invertible map in place of a Monte Carlo chain. [Noé et al., *Science* 2019]

- Source $\pi_0 \propto e^{-U_0}$, Gaussian. Normalizing flow G trained so that $G_{\#}\pi \approx \pi_0$.
- Pushforward $\nu := (G^{-1})_{\#}\pi_0$, tractable density $\nu(y) = \pi_0(G(y))|\det J_G(y)|$.
- Conventional losses: reverse and forward KL,

$$\text{(reverse)} \quad \text{KL}(\nu \parallel \pi) = \int_{\mathbb{R}^d} \nu(y) \log \frac{\nu(y)}{\pi(y)} dy,$$

$$\text{(forward)} \quad \text{KL}(\pi \parallel \nu) = \int_{\mathbb{R}^d} \pi(y) \log \frac{\pi(y)}{\nu(y)} dy.$$

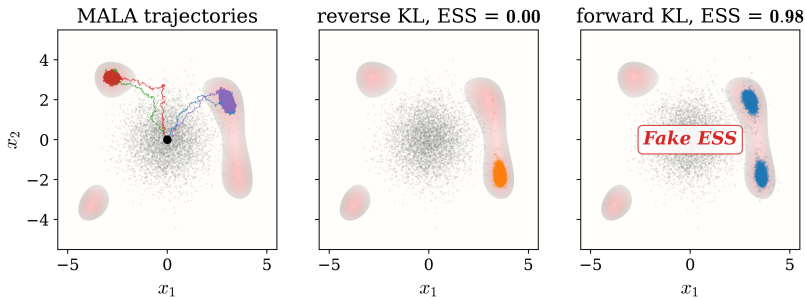
- Reverse KL: **silent** where ν misses a mode of π .
- Forward KL: needs Monte Carlo samples of π to train, and inherits their error.

Forward KL Can Still Have Mode Collapse

- Weight $w(y) = \pi(y)/\nu(y)$, log-ratio $z = \log w$:

$$z(y) = U_0(G(y)) - U(y) - \log|\det J_G(y)|, \quad \text{KL}(\pi\|\nu) = \mathbb{E}_{y\sim\pi}[z(y)].$$

- Effective sample size $\text{ESS}(w) = (\sum_i w_i)^2 / (N \sum_i w_i^2)$ does not guarantee mode coverage. It can be **fake**.



The Log-Ratio Variation

Wanted: a term added to the forward KL. Evaluated at points from any distribution we can generate, needs **no normalizing constant** of π , keeps $\nu = \pi$ a minimizer.

Log-ratio variation at a weighting distribution ω

$$X_{\omega}(\pi\|\nu) := \mathbb{E}_{y, y' \sim \omega} [|z(y) - z(y')|], \quad z = \log \frac{\pi}{\nu}.$$

Mean absolute pairwise difference of the log-ratio, both points drawn from ω .

- Normalizers cancel: a constant shift of z leaves every difference unchanged.
- $X_{\omega}(\pi\|\nu) = 0$ exactly when z is constant ω -a.s., i.e. $\nu \propto \pi$.
- ω says where flatness is demanded, and **it need not be π** . That freedom is the whole design.

The KLXX Loss

Two log-ratio variations appended to the forward KL, the two X's:

$$\begin{aligned}\text{KLXX}[G] &= \underbrace{\text{KL}(\pi\|\nu)}_{\text{target fit}} + \underbrace{X_{\pi}(\pi\|\nu)}_{\text{target variation}} + \underbrace{X_{(\hat{\pi}+\bar{\nu})/2}(\pi\|\nu)}_{\text{mixture variation}} \\ &= \mathbb{E}_{y\sim\pi}[z(y)] + \mathbb{E}_{y,y'\sim\pi}[|z(y) - z(y')|] + \mathbb{E}_{y,y'\sim(\hat{\pi}+\bar{\nu})/2}[|z(y) - z(y')|].\end{aligned}$$

- $\hat{\pi}$: the quench and temper (QT) distribution, supplying candidate modes.
- $\bar{\nu}$: the detached pushforward, equal to ν but outside the differentiation graph.

Generalized loss, mixture $\xi := \alpha\hat{\pi} + \beta\bar{\nu}$, with $\lambda, \vartheta, \alpha, \beta \geq 0$, $\alpha + \beta = 1$:

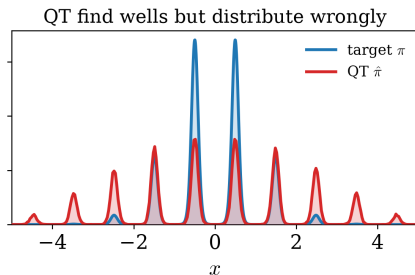
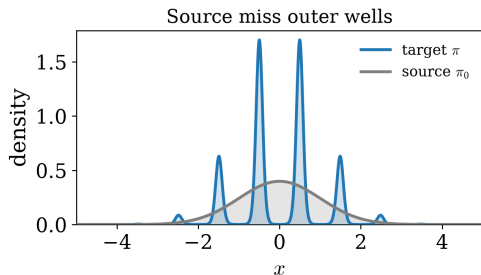
$$\mathcal{L}[G] = \text{KL}(\pi\|\nu) + \lambda X_{\pi}(\pi\|\nu) + \vartheta X_{\alpha\hat{\pi}+\beta\bar{\nu}}(\pi\|\nu).$$

KLXX is determined by $(\lambda, \vartheta, \alpha, \beta) = (1, 1, \frac{1}{2}, \frac{1}{2})$, written $\text{KL}+X_{\pi}+X_{(\hat{\pi}+\bar{\nu})/2}$.

Quench and Temper: Finding Candidate Modes

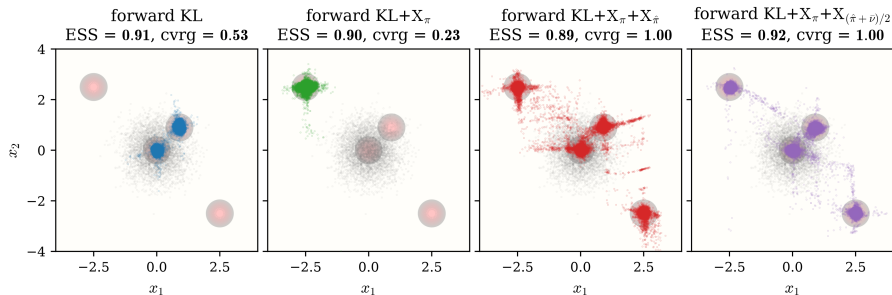
Three steps on each source sample $y \sim \pi_0$. Oracle access to U and ∇U only.

- 1 Melt: $y \leftarrow y + m_e \zeta$, $\zeta \sim \mathcal{N}(0, I_d)$, at melt scale $m_e > 0$.
- 2 Quench: $\dot{y} = -\nabla U(y)$, into whichever basin the melted particle landed in.
- 3 Temper: $\dot{y} = -\nabla U(y) + \sqrt{2} \dot{B}_t$, a spread around that basin.



Why Mix QT with the Pushforward

- $\hat{\pi}$ is inaccurate everywhere. Admissible: a variation asks for flatness of z , never for the mass at those points.
- $X_{\hat{\pi}}$ weights points inside the basins only, and says nothing where ν puts mass and $\hat{\pi}$ does not: the **intermodal** regions.



The Collision: Why Mixture Handles Collapse?

Under the **mixture** $\xi = \frac{1}{2}(\hat{\pi} + \bar{\nu})$, training drives

$$z(y) = \log \frac{\pi(y)}{\nu(y)} \approx \log \frac{\pi(y')}{\nu(y')} = z(y'), \quad y \sim \hat{\pi}, \quad y' \sim \bar{\nu}.$$

- y in a target mode: $z(y) \rightarrow +\infty$ if ν **misses** it.
- y' where the flow puts mass: $z(y') \rightarrow -\infty$ if ν **leaks**.
- Both hold only if ν covers every mode and avoids the intermodal regions.

KLXX Training Algorithm

Training of the generalized KLXX loss with parameters $(\lambda, \vartheta, \alpha, \beta)$.

Source samples $\mathcal{Y}_0 \sim \pi_0$ and QT samples $\hat{\mathcal{Y}} = \text{QT}(\mathcal{Y}_0, U)$, fixed before training.

Per gradient step, run:

- 1 pushforward batch $\mathcal{X}_\nu = G^{-1}(\mathcal{X}_0)$, from a random $\mathcal{X}_0 \subset \mathcal{Y}_0$ of size B ;
- 2 target surrogate $\mathcal{X}_\pi = \text{SMC}(\mathcal{X}_\nu)$, from pushforward ν to π ;
- 3 mixture batch $\mathcal{X}_\xi \sim \xi = \alpha\hat{\pi} + \beta\bar{\nu}$, drawn from $\hat{\mathcal{Y}}$ and $\mathcal{X}_\nu$.

$$\widehat{\mathcal{L}}[G] = \underbrace{\widehat{\mathbb{E}}_{y \in \mathcal{X}_\pi} [z(y)]}_{\text{forward KL}} + \underbrace{\lambda \widehat{\mathbb{E}}_{y, y' \in \mathcal{X}_\pi} [|z(y) - z(y')|]}_{\text{target variation}} + \underbrace{\vartheta \widehat{\mathbb{E}}_{y, y' \in \mathcal{X}_\xi} [|z(y) - z(y')|]}_{\text{mixture variation}}$$

One Adam step on $\widehat{\mathcal{L}}[G]$ updates G ; only z carries the gradient.

The Empirical Averages: Exact Calculation

- On a batch $\mathcal{X} = \{y_1, \dots, y_B\}$, the empirical averages are

$$\widehat{\mathbb{E}}_{y \in \mathcal{X}}[z(y)] := \frac{1}{B} \sum_{i=1}^B z(y_i),$$

$$\widehat{\mathbb{E}}_{y, y' \in \mathcal{X}}[|z(y) - z(y')|] := \frac{2}{B(B-1)} \sum_{1 \leq i < j \leq B} |z(y_i) - z(y_j)|.$$

- Each $z(y_i)$ is a scalar. Sorting $z_{(1)} \leq \dots \leq z_{(B)}$ gives the double sum exactly,

$$\sum_{i < j} |z(y_i) - z(y_j)| = \sum_{k=1}^B (2k - B - 1) z_{(k)},$$

at cost $O(B \log B)$: the $\binom{B}{2}$ pairs are never formed.

Adaptive Staging Boltzmann Generator: Interpolation

One flow from π_0 to π is not enough in high dimension. Interpolate the potentials and train one flow per [stage](#):

$$U_t = (1 - t)U_0 + tU, \quad \pi_t \propto e^{-U_t}, \quad 0 = t_0 < t_1 < \dots < t_K = 1.$$

- Stage k carries $\pi_{k-1} \rightarrow \pi_k$, a deformation small enough for one flow.
- The schedule cannot be fixed in advance: no target data, no knowledge of where the wells are. The difficulty is unknown until it is attempted.
- **Accept a stage only when its sample ESS clears a gate τ_v** , otherwise shrink the increment and retry.
- Easy interpolations take large stages, hard ones take small.

Adaptive Staging Boltzmann Generator: Algorithm

Sample set $\mathcal{Y}_0 \sim \pi_0$, $t_0 = 0$. At the stage $t_{k-1} \mapsto t_k$:

- 1 propose t_k by extrapolating the last accepted increment,

$$t_k = \min(1, t_{k-1} + \Gamma(t_{k-1} - t_{k-2}));$$

- 2 build the **QT set** $\hat{\mathcal{Y}}_k = \text{QT}(\mathcal{Y}_{k-1}, U_{t_k})$; (prebuilt before training)

- 3 train G_k on $\pi_{k-1} \rightarrow \pi_k$ by the KLXX training algorithm using $\hat{\mathcal{Y}}_k$;

- 4 weights $w_k = \exp(z_k)$ on $\mathcal{Y}_{k-1}$ pushed through G_k^{-1} .

If $\text{ESS}(w_k) < \tau_v$: shrink $t_k \leftarrow t_{k-1} + \gamma(t_k - t_{k-1})$ and repeat from 2.

Otherwise: resample, rejuvenate under U_{t_k} , and set $\mathcal{Y}_k$.

Defaults $\tau_v = 0.4$, $\Gamma = 1.5$, $\gamma = 0.7$. Output: the schedule $\{t_k\}$ and the stage maps $\{G_k\}$, which the generator replays on fresh source samples.

How the Error Propagates

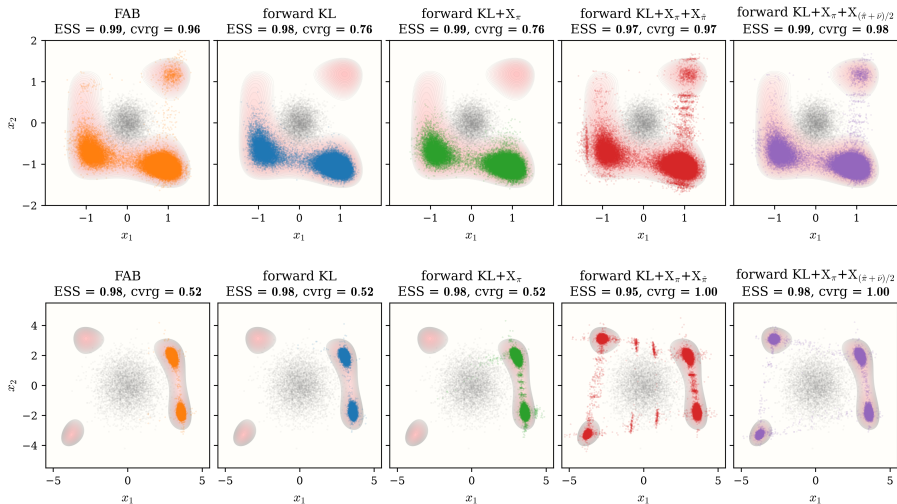
Inference replays the accepted stages on N fresh source particles: push forward, reweight, resample, rejuvenate.

- For every bounded observable φ , with π_K^N the returned empirical measure, $w_k = \pi_k / \nu_k$ the stage weights and $\text{osc}(\varphi) = \text{ess sup } \varphi - \text{ess inf } \varphi$,

$$\|\pi_K^N \langle \varphi \rangle - \pi_K \langle \varphi \rangle\|_2 \leq \frac{\text{osc}(\varphi)}{2\sqrt{N}} \sum_{k=0}^K \prod_{j=k+1}^K 3\|w_j\|_\infty,$$

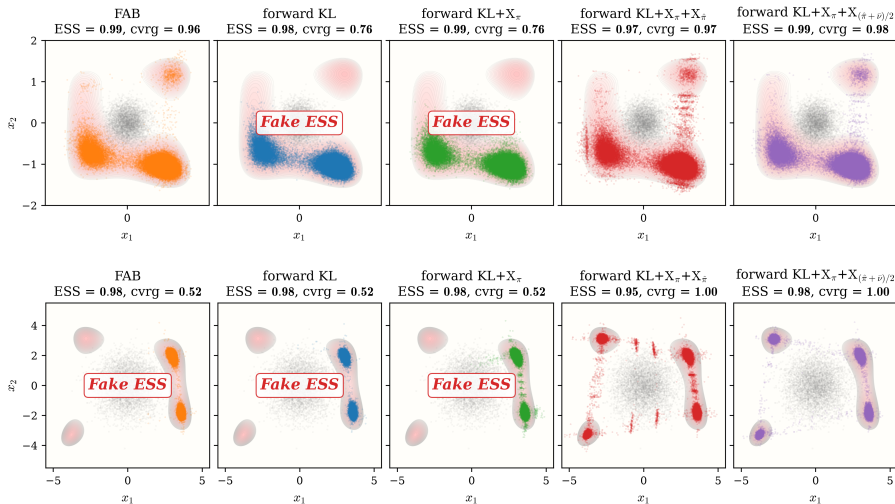
- The scheme is asymptotically unbiased: its error is removed by raising N alone.** Classical Monte Carlo instead needs the simulation time to grow.

2D Benchmarks: Three-Well and Himmelblau



Three-Well (top), Himmelblau (bottom). Every method reports a high ESS.

2D Benchmarks: Where the ESS Is Earned

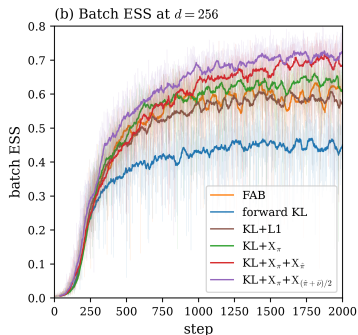
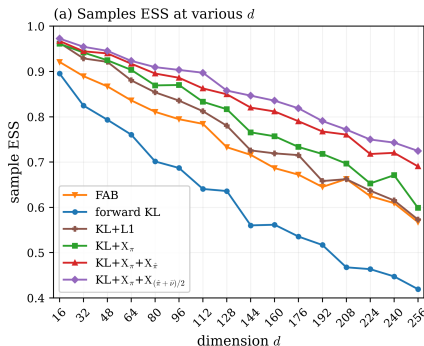


Coverage below 0.9: the ESS is earned only on the wells reached.

Product Multi-Well up to $d = 256$

$$U(\mathbf{x}) = \frac{1}{2}\|\mathbf{x}\|^2 + 12 \sum_{i=1}^k e^{-x_i^2}, \quad k = \lfloor \log_2 d \rfloor, \quad d = 16, \dots, 256.$$

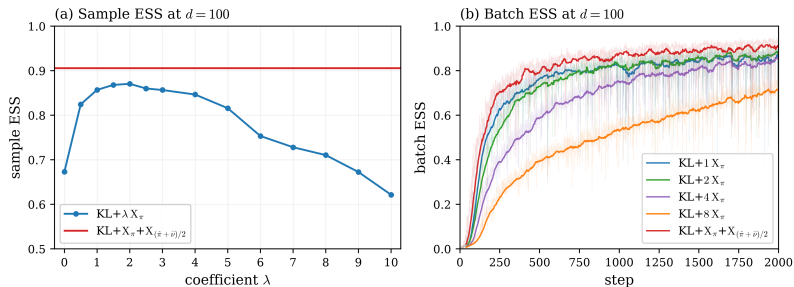
- KL+L1: log-dispersion regularization penalty LDR-L1 (Schopmans et al.)
- FAB: flow annealed importance sampling bootstrap (Midgley et al.)



Every log-ratio variation beats forward KL at all sixteen d .

Coefficient Sweep at $d = 100$

$\text{KL} + \lambda X_\pi$ on the same potential, $\lambda = 0$ being forward KL.



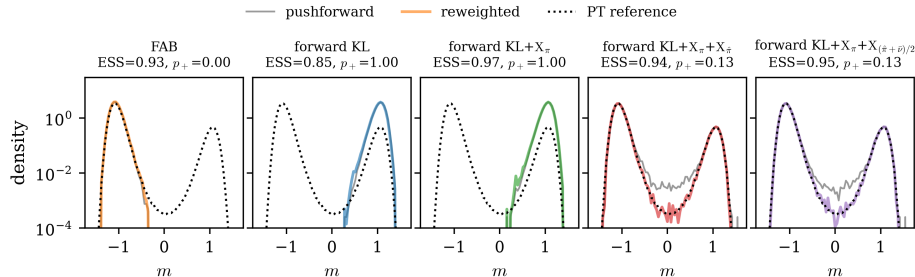
KLXX (0.906, red line) sits above every value the sweep attains. Tuning one coefficient does not reproduce the mixture variation.

Lattice ϕ^4 : a Collective Barrier

Tilted scalar ϕ^4 action on a periodic $L \times L$ lattice, $d = L^2$, at $L = 6$, $h = 0.0257$:

$$U(\phi) = -0.8 \sum_{\langle jl \rangle} \phi_j \phi_l + \sum_{j=1}^d \left[\phi_j^2 + \frac{1}{2} (\phi_j^2 - 1)^2 \right] + h \sum_{j=1}^d \phi_j, \quad m(\phi) = \frac{1}{d} \sum_{j=1}^d \phi_j.$$

A domain-wall barrier separates the two wells; the minority weight $p_+ = \mathbb{P}(m > 0)$ must be learned. Only $\hat{\pi}$ -based losses recover the minority weight.



Boltzmann Generator on Three Molecules

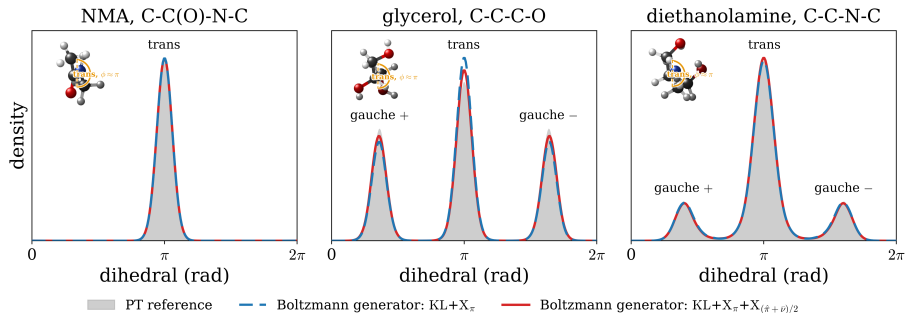
- The difficulty is not multimodality alone: the Lennard–Jones core is **singular**, so an early proposal can land at energies far outside the thermally occupied region.
- Implicit solvent at 300 K, no target data, one seed. The path is regularized, ending short of the bare potential.

molecule (d)	KL+X $_{\pi}$		KLXX	
	$\hat{F}$	stages	$\hat{F}$	stages
<i>N</i> -methylacetamide, NMA (30)	2.06	6	1.77	5
glycerol (36)	3.77	7	3.28	6
neutral diethanolamine (48)	19.90	10	15.84	10

- Propagation factor $\hat{F} = \prod_k \text{ESS}_k^{-1/2}$, from the sample ESS of each accepted stage; smaller is better.

Dihedral Marginals against Parallel Tempering

Reference: parallel tempering, six replicas from 300 to 800 K. Densities symmetrized under $\varphi \mapsto -\varphi$, as the achiral targets allow.



NMA shows one well: its gauche population is too small to appear. Largest departure: glycerol trans, KLXX at about half of the KL+X $_{\pi}$ excess.

Conclusions

- Forward KL trains only on the samples it is given. When a mode is missing from the flow and from those samples, the ESS stays high and says nothing: a **fake ESS**.
- X_ω compares $\log(\pi/\nu)$ at pairs of points. It needs no normalizing constant, and it can be evaluated wherever we can draw samples: at π for accuracy, at $\hat{\pi}$ for the modes the flow has not found, at $\bar{\nu}$ to keep mass out of the gaps.
- The loss itself makes the flow **visit every mode**. The extra cost per step is two sorts and one more batch, plus the QT set built once.

- F. Noé, S. Olsson, J. Köhler, H. Wu. Boltzmann generators. *Science* 365 (2019).
- L. Midgley, V. Stimper, G. Simm, B. Schölkopf, J. Hernández-Lobato. Flow annealed importance sampling bootstrap. *ICLR* (2023).
- H. Schopmans, C. von Klitzing, P. Friederich. Efficient training of Boltzmann generators using off-policy log-dispersion regularization. arXiv:2602.03729 (2026).
- C. Durkan, A. Bekasov, I. Murray, G. Papamakarios. Neural spline flows. *NeurIPS* (2019). The architecture used at every stage here.